

Digital Economy and Industrial Employment Upgrading in China

Evidence from Provincial Panel Data, 2013–2022

Airui Meng

Teachers College, Columbia University

Part I

Motivation, Research Questions, Data & Methods

Motivation: digitalization and where people work

- **A scene:** a decade ago you queued at the bank and waved down taxis on the street; today in Hangzhou or Shenzhen you pay, borrow, shop, and book a doctor through one phone app—while work in interior Gansu looks very different. *As provinces went digital, did the **kinds of jobs** people hold change?*
- Since the early 2010s, China's **digital economy**—broadband, mobile internet, e-commerce, digital payments—has expanded rapidly and reorganized how labor is allocated across sectors.
- Prior work blurs **two distinctions** this project keeps separate:
 - **Gross sectoral reallocation** (labor moving *into* services) vs. upgrading *within* services (toward producer / knowledge-intensive activities).
 - **Between-province** variation (rich coast vs. interior) vs. **within-province** change over time.
- A 2013–2022 provincial panel lets me ask: does digital development track employment *upgrading*—and *where* does that association live?

Research questions

Primary RQ

How is provincial **digital-economy development (DEDI)** associated with **industrial employment upgrading**—chiefly the *tertiary* employment share, with the primary and secondary shares as parallel outcomes—in China, 2013–2022?

Secondary RQ 1 (within-service composition)

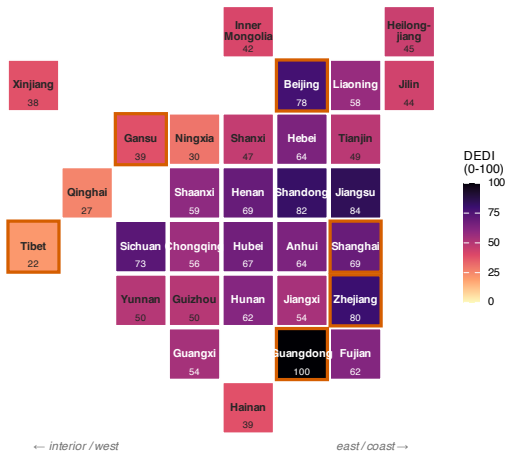
Within services, is digital development more strongly associated with *producer-service* employment (IT, finance, scientific research, business) than with *consumer-* or *public-service* employment?

Secondary RQ 2 (regional heterogeneity)

Does the association differ between *coastal* and *inland* provinces?

The setting: China's regional digital divide

One square per province (roughly by geography); darker = more digital (DEDI, 2022).
Orange outline = the provinces named elsewhere in this talk.



- China's digital economy is **deeply uneven across space**: the coastal east (Guangdong, Jiangsu, Zhejiang, Beijing) scores roughly **2–4×** the western interior (Tibet, Qinghai, Xinjiang).
 - This ordering is **large and persistent**—the ranking barely shifts from year to year.
- ⇒ That **between-province** gap is the backdrop for the whole talk: does a province's digital level track *what kinds of jobs* it has?

Data: from raw sources to regression variables

Sources: NBS yearbooks (Statistical, Labour, Tertiary Industry, Education, S&T) | PKU Digital Financial Inclusion Index | provincial communiqués.

Panel: 31 provinces $\times$ 10 years (2013–2022) = 310 province-years; balanced, 0 missing after cleaning.

	Raw measures		Construction		Final regression variable
Predictor	22 digital indicators (infrastructure, industry, environment)	$\Rightarrow$	PCA: 4 components, variance-weighted, rescaled to 0–100	$\Rightarrow$	DEDI (0–100; entered per 1 SD)
Outcomes	sector & service-group employment counts	$\Rightarrow$	$\div$ total employment (or classified service total)	$\Rightarrow$	tertiary / primary / secondary; producer / consumer / public / advanced shares
Controls	GDP, population, urbanization, trade, credit, fiscal	$\Rightarrow$	$\log$ (GDP, labor force) + z-standardize	$\Rightarrow$	six standardized controls + coastal indicator

Unit of analysis: the **province-year**. The advanced-service share is a subset of producer services.

Data and variables

- **Panel:** 31 provinces $\times$ 10 years = **310 province-years** (NBS yearbooks + PKU Digital Financial Inclusion Index + communiqués); balanced, 0 missing.
- **Predictor — DEDI:** a digital-economy development index from **PCA of 22 indicators** (digital infrastructure, industry, environment), rescaled to 0–100; reported **per 1 SD** (= 17.8 pts).
- **Outcomes (proportions):**
 - Broad structure: tertiary / primary / secondary employment shares.
 - Within services: producer / consumer / public / advanced shares (advanced = IT+finance+sci-research, a *subset* of producer).
- **Controls:** (log) GDP p.c., urbanization, (log) labor force, trade openness, financial development, government intervention; coastal indicator.
- Interpretation throughout is **associational**, not causal.

Variables and descriptive statistics ($N = 310$)

Continuous variables: M (SD); categorical: n (%).

Outcomes are proportions; DEDI entered per 1 SD (= 17.8 pts).

Industrial employment structure

Tertiary share	0.45 (0.10)
Secondary share	0.25 (0.09)
Primary share	0.29 (0.13)
Upgrading index (1–3)	2.15 (0.22)

Service-sector composition

Producer-service	0.28 (0.07)
Consumer-service	0.13 (0.05)
Public-service	0.54 (0.12)
Advanced ($\subset$ producer)	0.15 (0.04)

Digital economy (predictor)

DEDI (0–100)	48.7 (17.8)
--------------	-------------

Economic & institutional controls

GDP per capita (10k CNY)	6.38 (3.07)
Urbanization rate	0.60 (0.12)
Labor force (log)	7.51 (0.87)
Trade openness	0.25 (0.26)
Financial development	1.55 (0.44)
Government intervention	0.29 (0.20)

Location (n , %)

Coastal	120 (39)
Inland	190 (61)

Service shares use the classified urban-unit service base (4 groups: producer, consumer, public, mixed [not shown]); advanced $\subset$ producer. Coastal = 12 sea-bordering provinces. Gov. intervention = fiscal expenditure/GDP; financial development = loans/GDP; upgrading index = $1*\text{prim}+2*\text{sec}+3*\text{tert}$.

Model 1 — two-way fixed-effects OLS (the workhorse)

$$Y_{it} = \beta_0 + \beta_1 \text{DEDI}_{it} + \theta' \mathbf{X}_{it} + \underbrace{\alpha_i}_{\text{province FE}} + \underbrace{\lambda_t}_{\text{year FE}} + \underbrace{\varepsilon_{it}}_{\text{idiosyncratic}}$$

- **Idea:** compare each province *to its own past*— α_i nets out fixed local traits (geography, institutions), λ_t nationwide year shocks (policy, business cycle)—so β_1 is the **within-province** association per 1 SD of DEDI. $\mathbf{X}_{it}$ = the six standardized controls (θ their coefficients).
- **Assumptions:**
 - *Strict exogeneity:* $E[\varepsilon_{it} \mid \text{DEDI}_{it}, \mathbf{X}_{it}, \alpha_i, \lambda_t] = 0$ — plainly, nothing left in the error moves with DEDI. The key (untestable) assumption $\Rightarrow$ read *associationally*.
 - Errors are *not* i.i.d.: heteroskedastic and serially correlated within province, independent across $\Rightarrow$ **SE clustered by province**.
 - Linearity in parameters; no perfect collinearity.
- **Maps to the RQs:** Primary = broad shares; RQ 1 = refit with service shares; RQ 2 = add $\beta_2 (\text{DEDI}_{it} \times \text{Coastal}_i)$.
- DEDI is $\sim 84\%$ *between-province* $\Rightarrow$ FE isolates a thin within signal (conservative); a **Hausman / Mundlak** test (does the province effect correlate with DEDI?) favours fixed over random effects.

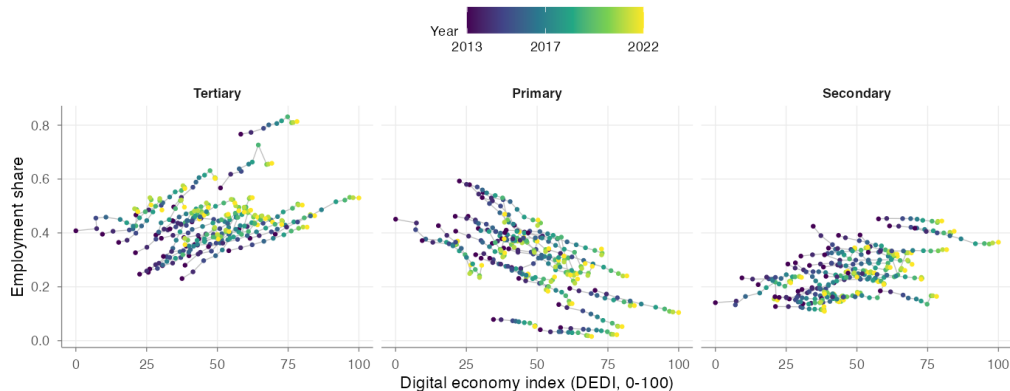
Model 2 — multilevel growth model (descriptive extension)

$$Y_{it} = \gamma_0 + \gamma_1 \text{DEDI}_{it} + \gamma_2 \text{Year}_t + \boldsymbol{\theta}' \mathbf{X}_{it} + \underbrace{u_{0i}}_{\text{random int.}} + \underbrace{u_{1i} \text{Year}_t}_{\text{random slope}} + \varepsilon_{it}$$

$$\begin{pmatrix} u_{0i} \\ u_{1i} \end{pmatrix} \sim \mathcal{N}(\mathbf{0}, \mathbf{G}), \quad \mathbf{G} = \begin{pmatrix} \tau_0^2 & \tau_{01} \\ \tau_{01} & \tau_1^2 \end{pmatrix}, \quad \varepsilon_{it} \sim \mathcal{N}(0, \sigma^2)$$

- **Assumptions:** random effects are *bivariate normal* with an *unstructured* $\mathbf{G}$ (intercept and slope may correlate, $\rho_{01} = \tau_{01}/\tau_0\tau_1$); the residual is normal and *independent of the random effects* (conditional independence given u_{0i}, u_{1i}); year enters *linearly* (not as year FE); estimated by **REML**.
- **Variance components:** τ_0^2 = between-province variance in the *baseline*; τ_1^2 = variance in the *upgrading slope*; ρ_{01} = intercept–slope correlation; σ^2 = within-province residual variance.
- **Outputs:** $\text{ICC} = \tau_0^2/(\tau_0^2 + \sigma^2)$ (share of variance *between* provinces); the random slopes are province-specific baselines and trajectories—which provinces upgrade fastest.
- Unlike Model 1, it **pools between + within** variation (partial pooling) — so it is descriptive, and the FE vs. mixed contrast shows how much association is within vs. between.

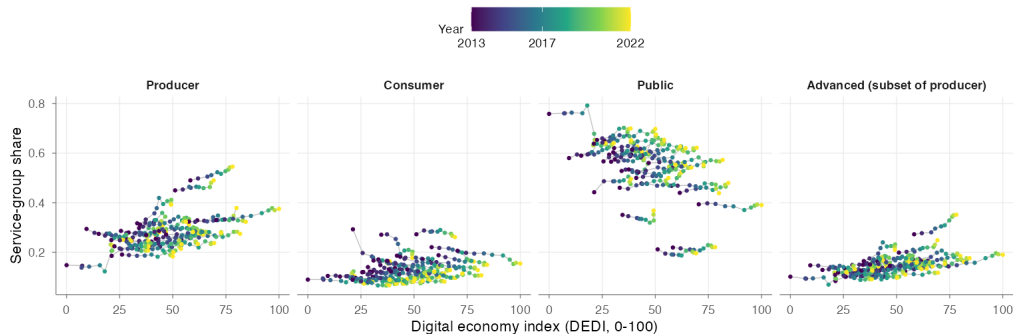
Plot 1 — Province trajectories: DEDI vs. broad sector shares



Each path = one province over 10 years (color = year); all 31 provinces; shared axes.

- **Primary RQ:** tertiary share **rises** and primary **falls** with DEDI; secondary is non-monotonic.
- **Motivates the design:** the wide *horizontal* spread is **between**-province; the short paths are **within**-province—the two sources Models 1–2 separate.

Plot 2 — Within services: DEDI vs. service-group shares



Same provinces and years, now *within* classified urban-unit service employment.

- **Secondary RQ 1:** producer and advanced shares **rise** with DEDI; consumer is roughly flat; public **falls**.
- Suggests a *selective upgrading* toward higher-value producer / advanced services—the hypothesis Model 1 will test under fixed effects.

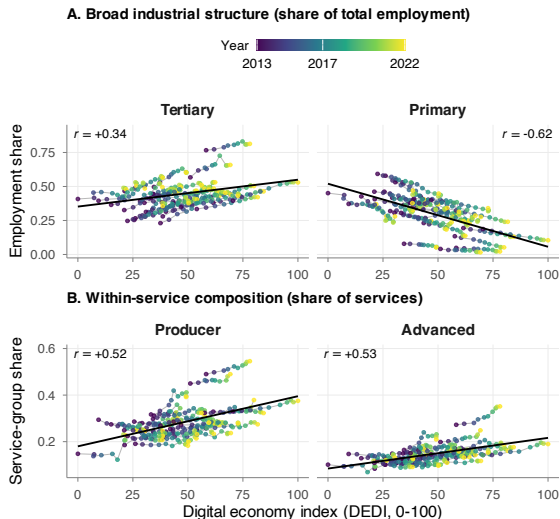
Part I summary — the setup

- **Set up:** a 310 province-year panel; a PCA-based digital index (DEDI); **two models** mapped to **one primary + two secondary** research questions; EDA that motivates the questions.
- **Expected contribution:** separate *gross sectoral reallocation* from *within-service upgrading*, and *between-* from *within*-province variation—under-examined for China at the provincial level.
- **Part II (next):** the fixed-effects estimates, the between-vs-within decomposition, and the multilevel results.

Part II

Results & Findings — what the data say

Cross-sectionally, the link looks strong...



- Pooled across all 310 province-years, DEDI is **positively** correlated with the tertiary share ($r = .34$) and the producer ($r = .52$) and advanced ($r = .53$) service shares, and **negatively** with the primary share ($r = -.62$).

- But each thin grey path is one province across 2013–2022 (dots coloured by year): the paths are short, while provinces sit far apart along DEDI — the spread is overwhelmingly **between** provinces (Guangdong vs. Tibet).

⇒ So is this a **within-province** relationship, or just that developed provinces are *both* more digital *and* more service-oriented?

...but it is *between* provinces, not *within*

Between vs. within: DEDI coefficient by outcome (per SD)

Broad shares: positive between, null within. Producer/advanced services: positive in BOTH.



Bars = DEDI coefficient per 1 SD; whiskers = 95% CI.

DEDI coefficient on the **tertiary share** (change in the share, per 1 SD), across estimators:

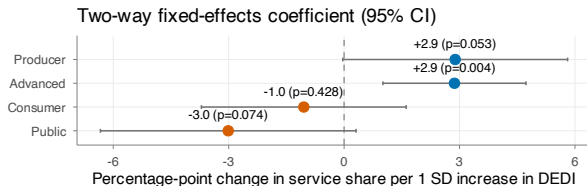
Estimator	Δ tertiary share
Pooled	+0.086**
Between-province	+0.067 ⁺
Random effects	+0.050*
One-way (province) FE	+0.047*
Two-way FE	-0.002

Significance: ⁺ $p < .10$, * $p < .05$, ** $p < .01$.

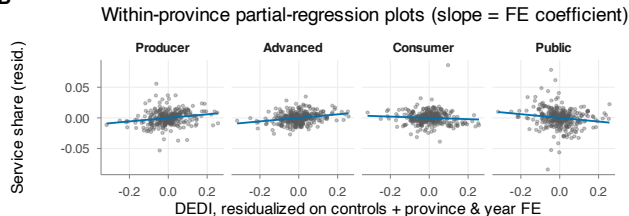
- Positive everywhere *until* year effects are added—then it **collapses to null**.
 - DEDI is ~84% between-province; province+year FE absorb ~99%. Hausman/Mundlak select FE; DEDI's between vs. within slopes are statistically **indistinguishable**.
- ⇒ The broad link is a **between-province** pattern (a nationwide trend over the decade), not a within-province effect.

The robust signal: upgrading *within* services

A



B

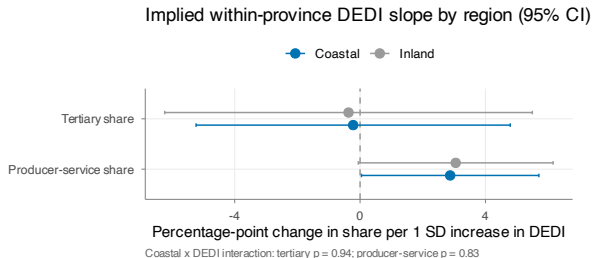


Panel A dots: blue = positive, orange = negative; bars = 95% CI.

- Even under two-way FE, a 1-SD rise in DEDI is associated with a higher **advanced-service share** (+2.9 pp, $p = .004$).
 - Producer-service points the same way but is borderline; public ↓; consumer flat.
- ⇒ Digital development tracks **compositional upgrading** toward high-value services.

No coastal–inland difference (RQ 2)

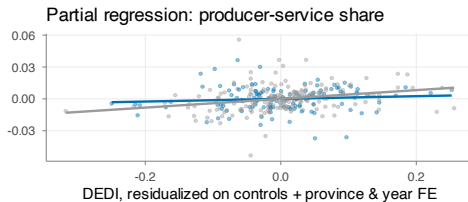
A



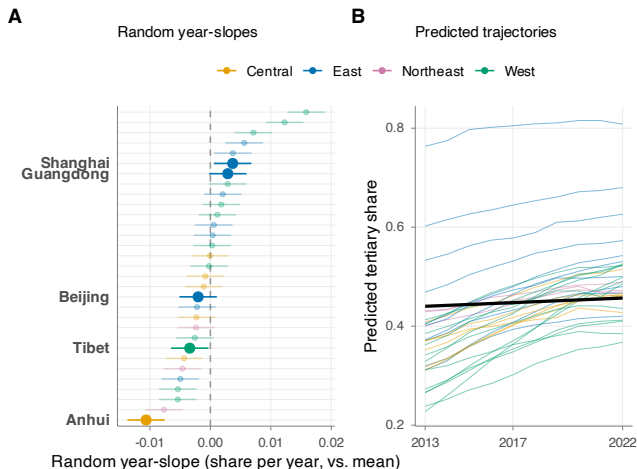
- The implied within-province DEDI slopes are nearly identical for coastal and inland provinces.
- Interaction not significant: $p = .94$ (tertiary), $p = .83$ (producer).

⇒ The digital–services link is **not** a coastal phenomenon.

B
Producer-service share (resid.)



Heterogeneous upgrading trajectories (Model 2)



- $ICC = 0.86$ (tertiary) / 0.98 (producer): most variation is *between* provinces—echoing the FE story.
- Random year-slopes reveal which provinces upgrade fastest; intercept-slope correlation -0.71 (higher-baseline provinces upgrade more slowly).
- Producer-service is positive across FE and both multilevel models.

What might explain it — and one caveat

Plausible mechanisms (interpretive, *not* causal):



- **Structural caveat:** DEDI and advanced services *overlap conceptually*—DEDI partly captures the same IT and digital-finance activity the advanced-service share measures.
- ⇒ Best read as a robust **descriptive regularity of the knowledge economy**, not an effect of digital infrastructure on service composition.

Conclusion, contribution, and limitations

Digital economy → not *more* services, but *better* services —
and that only shows once you separate *between-* from *within-province* variation.

- **Bottom line:** the digital economy is linked less to *gross sectoral reallocation* than to *compositional upgrading within services*, toward higher-value advanced (and, more weakly, producer) activities.
- **Contribution:** separating *between* from *within* reconciles the large cross-sectional correlations with the muted within-province estimates—and corrects a common conflation.
- **Limitations:** associational (FE removes time-invariant, not time-varying confounders or reverse causality); the within-province DEDI signal is thin and collinear; outcomes use the classified urban-unit base; $N = 310$.

Thank you.